

AEO «Nazarbayev intellectual schools»
Center of pedagogical measurements



EXAMPLE CANDIDATE RESPONSE OF EXTERNAL SUMMATIVE ASSESSMENT

12 grade

CHEMISTRY
Paper 3

Astana, 2023 year

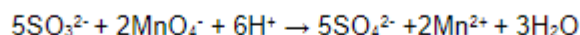
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QUESTION 1(a), (b) (i), (ii)

- 1 In this experiment you will find the percentage purity of a sample of sodium sulfite, Na_2SO_3 , by titration with acidified potassium manganate(VII), KMnO_4 .

The ionic equation for this reaction is shown.



- (a) Container A contains impure sodium sulfite, Na_2SO_3 .

Weigh container A and its contents. Record this value.

Pour the impure sodium sulfite into a 250 cm^3 beaker.

Weigh empty container A. Record this value.

Calculate the mass of impure sodium sulfite transferred to the beaker.

Record these mass readings in an appropriate table.

[2]

- (b) Add about 100 cm^3 of distilled water to the beaker. Stir the mixture with a stirring rod until all of the sodium sulfite has dissolved.

Transfer this solution into a 250 cm^3 graduated flask. Wash the beaker with small amounts of distilled water and transfer this water into the graduated flask.

Then fill the graduated flask up to the mark with distilled water.

Put a stopper on the graduated flask and thoroughly shake the mixture. Label this solution B.

Fill a burette with solution B. Record the initial reading of solution B in the burette.

Transfer 25.0 cm^3 of the $0.0200\text{ mol dm}^{-3}$ $\text{KMnO}_4(\text{aq})$ into a 250 cm^3 conical flask using a pipette.

Use a measuring cylinder to add 15.0 cm^3 of 1.00 mol dm^{-3} sulfuric acid to the conical flask.

Titrate the 25.0 cm^3 of $\text{KMnO}_4(\text{aq})$ against solution B. The end-point is the first appearance of a persistent colourless solution.

Record the final reading on the burette and determine the volume (titre) of solution B added.

Repeat until you obtain concordant results.

- (i) Record the results from the titrations in this table. You do not need to use all columns.

titration number						
initial (burette) reading / cm³						
final (burette) reading / cm³						
volume of solution B added / cm³						

[7]

- (ii) Calculate the mean volume of solution **B** added.

State which titration numbers you used to calculate this mean volume.

Give your answer to **two** decimal places.

titration numbers			
mean volume		cm ³	[2]

MARK SCHEME

Question	Answer	Mark	Additional guidance
1(a)	table showing mass of container and sample, mass of container and mass of sample with correct units masses recorded from correct calculation of mass of sample and all to the same number of decimal places (either 2 or 3)	1 1 [2]	all units must be shown in the headings ignore units elsewhere in the table accept results table with rows and columns reversed and ruled lines not necessary
1(b)(i)	final and initial burette readings correctly entered are all given to two decimal places with the last number being a 0 or a 5 correct calculation to get volume of added solution – it must have 1 or 2 decimal places Accuracy scaled volume within ± 0.2 of Supervisor's volume (5) scaled volume within ± 0.3 of Supervisor's volume (4) scaled volume within ± 0.4 of Supervisor's volume (3) scaled volume within ± 0.5 of Supervisor's volume (2) scaled volume within ± 0.6 of Supervisor's volume (1)	1 1 max 5 [7]	 calculate scaled volume using the formula candidate mean volume x supervisor mass/candidate mass
1(b)(ii)	correct numbers calculated to either 2 or 3 decimal places using results that are within ± 0.20 correct answer	1 1 [2]	accept mean quoted to 1, 2 or 3 decimal places correctly rounded

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

- 1 In this experiment you will find the percentage purity of a sample of sodium sulfite, Na_2SO_3 , by titration with acidified potassium manganate(VII), KMnO_4 .

The ionic equation for this reaction is shown.



- (a) Container A contains impure sodium sulfite, Na_2SO_3 .

Weigh container A and its contents. Record this value.

Pour the impure sodium sulfite into a 250 cm^3 beaker.

Weigh empty container A. Record this value.

Calculate the mass of impure sodium sulfite transferred to the beaker.

Record these mass readings in an appropriate table.

mass of container A and its contents / g	mass of empty container A / g	mass of impure Na_2SO_3 / g
16.24	13.29	2.95

[2]

- (i) Record the results from the titrations in this table. You do not need to use all columns.

titration number	rough	1	2	3		
initial (burette) reading / cm^3	0.00	15.00	0.00	15.00		
final (burette) reading / cm^3	15.00	30.00	15.00	30.00		
volume of solution B added / cm^3	15.00	15.00	15.00	15.00		

[7]

- (ii) Calculate the mean volume of solution B added.

State which titration numbers you used to calculate this mean volume.

Give your answer to **two** decimal places.

titration numbers 1, 2, 3 ✓
 mean volume $(15 + 15 + 15) : 3 = 15.00$ ✓

cm^3 [2]

EXAMINER'S COMMENTS

The candidate received 11 marks.

In question (a), the candidate earned two marks because he/she recorded the data in an appropriate table with the correct units. Additionally, the masses were determined from correct calculations, and the answers were written to the same decimal place.

In question (b)(i), the candidate received seven marks because he/she accurately entered data into the table with the last number being either a 0 or a 5. The candidate also calculated the volume of the solution with the correct decimal place, resulting in the candidate receiving the full

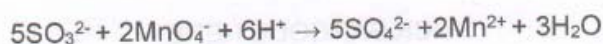
five marks for accuracy.

In question (b)(ii), the candidate obtained two marks for correctly using data from the table to calculate the mean volume and entering the answer correctly rounded.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

- 1 In this experiment you will find the percentage purity of a sample of sodium sulfite, Na_2SO_3 , by titration with acidified potassium manganate(VII), KMnO_4 .

The ionic equation for this reaction is shown.



- (a) Container A contains impure sodium sulfite, Na_2SO_3 .

Weigh container A and its contents. Record this value.

Pour the impure sodium sulfite into a 250 cm^3 beaker.

Weigh empty container A. Record this value.

Calculate the mass of impure sodium sulfite transferred to the beaker.

Record these mass readings in an appropriate table.

N	Items	mass (g)
1	Container A	32,51 g
2	content	301 g
3	Container with a cont	35,52 g

$$m(\text{content}) = 35,52\text{g} - 32,51\text{g} = 3,01\text{g}$$

[2]

- (i) Record the results from the titrations in this table. You do not need to use all columns.

titration number	1	2	3			
initial (burette) reading / cm^3	0,00	0,00	0,00			
final (burette) reading / cm^3	14,6	15,1	15,5			
volume of solution B added / cm^3	14,6	15,1	15,5			

[7]

- (ii) Calculate the mean volume of solution B added.

State which titration numbers you used to calculate this mean volume.

Give your answer to **two** decimal places.

titration numbers 1 and 2

mean volume $\frac{14,6 + 15,1}{2} = 14,85$ cm^3 [2]

EXAMINER'S COMMENTS

This candidate received seven marks out of eleven.

In question (a), the candidate recorded an appropriate table with correct units. Additionally, the masses were calculated correctly, and the answer was written to the same decimal place.

In question (b)(i), the candidate received four marks because of incorrectly entered data in the table, with the last number not being a 0 or a 5. The candidate also miscalculated the volume of the solution with an incorrect decimal place. Consequently, the candidate received three marks out of five for accuracy.

In question (b)(ii), the candidate lost one mark due to incorrect usage of data from the table to calculate the mean volume. However, the candidate entered the answer correctly rounded.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

- 1 In this experiment you will find the percentage purity of a sample of sodium sulfite, Na_2SO_3 , by titration with acidified potassium manganate(VII), KMnO_4 .

The ionic equation for this reaction is shown.



- (a) Container A contains impure sodium sulfite, Na_2SO_3 .

Weigh container A and its contents. Record this value.

Pour the impure sodium sulfite into a 250 cm^3 beaker.

Weigh empty container A. Record this value.

Calculate the mass of impure sodium sulfite transferred to the beaker.

Record these mass readings in an appropriate table.

Initial weigh of container A with content/g	17.015
Final weigh of container A without content /g	14.043
Mass of content/g	2.972

[2]

- (i) Record the results from the titrations in this table. You do not need to use all columns.

titration number	1	2	3	4		
initial (burette) reading / cm^3	0.00	10.00	0.00	16.00		
final (burette) reading / cm^3	16.30	26.40	16.10	32.10	✓	
volume of solution B added / cm^3	16.30	16.40	16.10	16.10		

[7]

- (ii) Calculate the mean volume of solution B added.

State which titration numbers you used to calculate this mean volume.

Give your answer to **two** decimal places.

titration numbers 1 (16.30), 3 (16.10), 4 (16.10) ✓

mean volume 16.17 ✓ cm^3 [2]

EXAMINER'S COMMENTS

This candidate received four marks. A candidate clearly demonstrated knowledge about isotopes in part a (i,ii,iii). In (i) question, the candidate used the key word 'different neutron number'.

In (ii), the candidate identified isotope and wrote its symbol.

In (iii), the candidate correctly used data in the table to calculate the relative atomic mass of copper, also the answer was written in one decimal place.

EXAMINER'S COMMENTS

This candidate received six marks out of eleven.

In question (a), the candidate earned two marks for recording an appropriate table with correct units. Additionally, the masses were calculated correctly, and the answer was written to the same decimal place.

In question b(i), the candidate correctly entered data into the table with the last number being a 0 or a 5. The candidate also calculated the volume of the solution with the correct decimal place. However, the candidate did not receive any marks for accuracy.

In question b(ii), the candidate correctly used data from the table to calculate the mean volume and entered the answer correctly rounded.

QUESTION 1c(i), (ii), (iii), (iv)

- (c) (i)** Calculate the amount, in moles, of KMnO_4 in 25.0 cm^3 of $0.0200\text{ mol dm}^{-3}$ KMnO_4 solution.

	mol	[1]
--	-----	-----

- (ii)** Calculate the amount, in moles, of Na_2SO_3 in the mean volume of solution **B** added.

Use your answer to **(c)(i)** and the equation for the reaction.

	mol	[1]
--	-----	-----

- (iii)** Calculate the amount, in moles, of Na_2SO_3 in the 250 cm^3 of solution **B**.

	mol	[1]
--	-----	-----

- (iv)** Hence, calculate the mass of Na_2SO_3 in 250 cm^3 of solution **B**.

	g	[1]
--	---	-----

MARK SCHEME

1(c)(i)	moles of $\text{KMnO}_4 = 0.0005$ mole	[1]	
1(c)(ii)	moles of $\text{Na}_2\text{SO}_3 = 0.00125$ mole	[1]	
1(c)(iii)	$\frac{250}{\text{mean volume}} \times \text{c(ii)}$	[1]	ecf from c(ii)
1(c)(iv)	$1(c)(iii) \times 126$	[1]	ecf from c(iii)

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

(c) (i) Calculate the amount, in moles, of KMnO_4 in 25.0 cm^3 of $0.0200 \text{ mol dm}^{-3}$ KMnO_4 solution.

$$n = C \cdot V = \frac{25}{1000} \cdot 0.0200 = 0.0005 \text{ mol.}$$

0.0005 ✓ mol [1]

(ii) Calculate the amount, in moles, of Na_2SO_3 in the mean volume of solution B added.

Use your answer to (c)(i) and the equation for the reaction.

$$\begin{array}{r} 5 - 2 \\ x - 0.0005 \end{array}$$

$$x = 1.25 \cdot 10^{-3} = 0.00125 \text{ mol}$$

0.00125 ✓ mol [1]

(iii) Calculate the amount, in moles, of Na_2SO_3 in the 250 cm^3 of solution B.

$$\begin{array}{r} 0.00125 - 15 \\ x - 250 \end{array}$$

$$x = 0.0208$$

0.0208 ✓ mol [1]

(iv) Hence, calculate the mass of Na_2SO_3 in 250 cm^3 of solution B.

$$m(\text{Na}_2\text{SO}_3) = n \cdot M = 0.0208 \cdot (23 \cdot 2 + 32 + 48) = 2.621$$

2.621 ✓ g [1]

EXAMINER'S COMMENTS

The candidate received four marks out of four. The candidate clearly demonstrated knowledge about the amounts of substances in part a (i, ii, iii, iv). In question 1c (i), the candidate correctly calculated the amount in moles of potassium manganate (VII). In question 1c(ii), the candidate correctly identified the ratio of the solutions using the equation to calculate the amount in moles of sodium sulfite. In question 1c(iii), the candidate correctly calculated the amount in moles in 250 cm^3 . In question 1c(iv), the candidate correctly used the molecular mass and calculated the mass of sodium sulfite.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

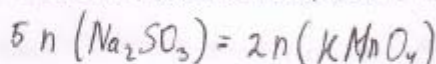
- (c) (i) Calculate the amount, in moles, of KMnO_4 in 25.0 cm^3 of $0.0200 \text{ mol dm}^{-3}$ KMnO_4 solution.

$$n(\text{KMnO}_4) = \frac{25}{1000} \times 0.0200 = 5 \times 10^{-4} \text{ mol}$$

$$5 \times 10^{-4} \text{ mol [1]}$$

- (ii) Calculate the amount, in moles, of Na_2SO_3 in the mean volume of solution B added.

Use your answer to (c)(i) and the equation for the reaction.



$$n(\text{Na}_2\text{SO}_3) = \frac{2 n(\text{KMnO}_4)}{5} = \frac{2 \times 10^{-4} \times 5.913 \times 10^{-4}}{5} \text{ mol [1]}$$

- (iii) Calculate the amount, in moles, of Na_2SO_3 in the 250 cm^3 of solution B.

$$n(\text{Na}_2\text{SO}_3) = \frac{25}{250} \times \frac{5.913 \times 10^{-4}}{2 \times 10^{-4}} = \frac{5.913 \times 10^{-4}}{2 \times 10^{-4}} = 2.9565 \times 10^{-3} \text{ mol [1]}$$

- (iv) Hence, calculate the mass of Na_2SO_3 in 250 cm^3 of solution B.

$$n = \frac{m}{M_r} \quad m(\text{Na}_2\text{SO}_3) = n(\text{Na}_2\text{SO}_3) \times M_r(\text{Na}_2\text{SO}_3) = 2.9565 \times 10^{-3} \times 126.1 = 0.373 \text{ g [1]}$$

EXAMINER'S COMMENTS

The candidate received two marks.

In question 1c (i), the candidate correctly calculated the amount in moles of potassium manganate (VII).

In question 1c(ii), the candidate incorrectly identified the ratio of the solutions while using the equation to calculate the amount in moles of sodium sulfite.

In question 1c(iii), the candidate used the incorrect volume of sodium sulfite to calculate the amount in moles in 250 cm^3 .

In question 1c(iv), the candidate correctly used the molecular mass and calculated the mass of sodium sulfite.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

- (c) (i) Calculate the amount, in moles, of KMnO_4 in 25.0 cm^3 of $0.0200 \text{ mol dm}^{-3}$ KMnO_4 solution.

$$n = c \cdot V_{\text{solution}}$$

$$n = 0.02 \cdot \frac{25}{1000} = 0.000328 \quad \dots\dots\dots 0.000328 \text{ or } 3.28 \cdot 10^{-4} \text{ mol [1]}$$

- (ii) Calculate the amount, in moles, of Na_2SO_3 in the mean volume of solution B added.

Use your answer to (c)(i) and the equation for the reaction.

$$n(\text{MnO}_4^-) : n(\text{SO}_3^{2-}) = 2 : 5$$

$$n(\text{Na}_2\text{SO}_3) = \frac{0.000328}{2} \cdot 5 = \dots\dots\dots 0.000820 \text{ or } 8.20 \cdot 10^{-4} \text{ mol [1]}$$

- (iii) Calculate the amount, in moles, of Na_2SO_3 in the 250 cm^3 of solution B.

$$25 \text{ cm}^3 - 8.2 \cdot 10^{-4} \text{ mol}$$

$$250 \text{ cm}^3 - x$$

$$x = \frac{250 \cdot 8.2 \cdot 10^{-4}}{25} = 8.20 \cdot 10^{-3} \quad \dots\dots\dots 0.00820 \text{ or } 8.20 \cdot 10^{-3} \text{ mol [1]}$$

- (iv) Hence, calculate the mass of Na_2SO_3 in 250 cm^3 of solution B.

$$m = n \cdot M$$

$$m = 0.00820_{\text{mol}} \cdot 126 \text{ g mol}^{-1} = 1.03 \text{ g [1]}$$

EXAMINER'S COMMENTS

The candidate received one mark.

In question 1c (i), the candidate incorrectly calculated the amount in moles of potassium manganate (VII).

In question 1c(ii), the candidate incorrectly identified the ratio of the solutions while using the equation to calculate the amount in moles of sodium sulfite.

In question 1c(iii), the candidate used the incorrect volume of sodium sulfite to calculate the amount in moles in 250 cm^3 .

In question 1c(iv), the candidate correctly used the molecular mass and calculated the mass of sodium sulfite.

QUESTION 1 d(i),(ii)

(d) (i) Calculate the percentage purity of the Na_2SO_3 .

Show your working.

	%	[1]
--	---	-----

(ii) Uncertainty of measurement of a burette is $\pm 0.05 \text{ cm}^3$.

Calculate the percentage uncertainty in the mean volume of **B** added during the titration.

Show your working.

	%	[2]
--	---	-----

MARK SCHEME

1(d)(i)	percentage purity = $\left(\frac{\text{(c)(iv)}}{\text{(mass of Na}_2\text{SO}_3 \text{ measured)}} \right) \times 100\%$ correct percentage purity calculated	1 1 [2]	ecf from c(iv)
(d)(ii)	$\frac{0.05 \times 2}{\text{mean volume}} \times 100\%$ correct percentage uncertainty calculated	1 1 [2]	ecf

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

(d) (i) Calculate the percentage purity of the Na_2SO_3 .

Show your working.

$$\frac{1,314 \text{ g} \cdot 100\%}{3 \text{ g}} = 43,8\%$$

..... 43,8 % [1]

For
Examiner
Use

2

(ii) Uncertainty of measurement of a burette is $\pm 0,05 \text{ cm}^3$.

Calculate the percentage uncertainty in the mean volume of B added during the titration.

$$V = 15,42 \text{ cm}^3$$

Show your working.

$$0,05 \text{ cm}^3 \cdot 2 = 0,1 \text{ cm}^3 \leftarrow \text{error}$$

$$\frac{0,10 \cdot 100\%}{15,42} = 0,649\%$$

$$\frac{15,42 \text{ cm}^3 - 100\%}{0,10 \text{ cm}^3} = ?$$

..... 0,649 % [2]

2

EXAMINER'S COMMENTS

The candidate received four marks out of four.

In question (d)(i), the candidate used the correct expression to calculate the purity of sodium sulfite and also correctly calculated the percentage.

In question (d)(ii), the candidate used the correct expression to calculate the percentage uncertainty and also correctly calculated the percentage.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

(d) (i) Calculate the percentage purity of the Na_2SO_3 .

Show your working.

$$m_0 = 2,9 \text{ g}$$

$$m(\text{Na}_2\text{SO}_3) = 2,52 \text{ g} \Rightarrow$$

$$\% = \frac{2,52}{2,9} \cdot 100\% = 86,9\%$$

..... 86,9 % [1]

For
Examiner's
Use

2

(ii) Uncertainty of measurement of a burette is $\pm 0,05 \text{ cm}^3$.

Calculate the percentage uncertainty in the mean volume of B added during the titration.

Show your working.

$$\% = \frac{0,05}{15,57} \cdot 100\% = 0,32\%$$

..... 0,32 % [2]

1

EXAMINER'S COMMENTS

The candidate received three marks.

In question 1d (i), the candidate used the correct expression to calculate the purity of sodium sulfite and correctly calculated the percentage.

In question 1d(ii), the candidate used an incorrect expression to calculate the percentage uncertainty; however, they correctly calculated the percentage.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

(d) (i) Calculate the percentage purity of the Na_2SO_3 .

Show your working.

$$\text{purity} = \frac{m(\text{calculated})}{m(\text{weighed})} \cdot 100\%$$

$$\text{purity} = \frac{1.03 \text{ g}}{3.08 \text{ g}} \cdot 100\% = 34.2\% \quad [1]$$

(ii) Uncertainty of measurement of a burette is $\pm 0.05 \text{ cm}^3$.

Calculate the percentage uncertainty in the mean volume of B added during the titration.

Show your working.

$$\left(\frac{0.05}{16.5} + \frac{0.05}{16.4} + \frac{0.05}{16.3} \right) \cdot 100\% = 0.915\%$$

$$0.915\% \quad [2]$$

For
Examiner
Use

2

0

EXAMINER'S COMMENTS

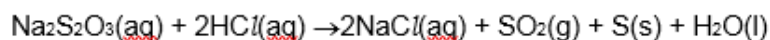
The candidate received two marks.

In question 1d (i), the candidate used the correct expression to calculate the purity of sodium sulfite and correctly calculated the percentage.

In question 1d (ii), the candidate lost one mark for using an incorrect expression to calculate the percentage uncertainty, and they also incorrectly calculated the percentage.

QUESTION 2 a (i)

- 2 In this experiment you will determine the order of reaction with respect to sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$, for the reaction with hydrochloric acid represented by the equation shown.



- (a) (i) Carry out the experiments and record your results in Table 2.1. All times recorded in whole numbers **and** in seconds.

Step 1. Draw the letter X on a piece of white paper.

Step 2. Use a large measuring cylinder to transfer 40.0 cm^3 of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ into a 100 cm^3 beaker.

Step 3. Use a clean large measuring cylinder to transfer 10.0 cm^3 of $\text{H}_2\text{O}(\text{l})$ into the beaker.

Step 4. Place the beaker over the piece of white paper with the letter X on it.

Step 5. Use a 10.0 cm^3 measuring cylinder to measure 5.0 cm^3 of $\text{HCl}(\text{aq})$.

Step 6. Pour the 5.0 cm^3 of $\text{HCl}(\text{aq})$ into the beaker **and** at the same time start a stop-watch.

Step 7. Immediately stir the mixture for 3 seconds.

Step 8. Record the time, to the nearest second, when the letter X is no longer visible when looking from above the beaker.

Step 9. Empty the mixture in the beaker labelled waste and thoroughly rinse the reaction beaker with water.

Step 10. Dry the beaker with a paper towel.

Repeat steps 2 to 10 with different volumes of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ as shown in Table 2.1.

Table 2.1

experiment number	volume of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ / cm^3	volume of $\text{H}_2\text{O}(\text{l})$ / cm^3	volume of $\text{HCl}(\text{aq})$ / cm^3	time / s	rate of reaction / s^{-1}
1	40.0	10.0	5.0		
2	30.0	20.0	5.0		
3	25.0	25.0	5.0		
4	20.0	30.0	5.0		
5	15.0	35.0	5.0		
6	10.0	40.0	5.0		

MARK SCHEME

2a(i)	Time values are increasing	1	
	All times recorded as whole numbers	1	
	Accuracy		
	Any three times within ± 3 s of Supervisors results (3)		
	Any two times within ± 6 s of Supervisors results (2)		
	Any one times within ± 6 s of Supervisors results (1)	[max3]	
		[5]	

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

Repeat steps 2 to 10 with different volumes of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ as shown in Table 2.1.

Table 2.1

experiment number	volume of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ / cm^3	volume of $\text{H}_2\text{O}(\text{l})$ / cm^3	volume of $\text{HCl}(\text{aq})$ / cm^3	time / s	rate of reaction / s^{-1}
1	40.0	10.0	5.0	22	48
2	30.0	20.0	5.0	28	36
3	25.0	25.0	5.0	33	30
4	20.0	30.0	5.0	41	24
5	15.0	35.0	5.0	59	17
6	10.0	40.0	5.0	92	11

✓✓ ✓✓✓

[5]

5

EXAMINER'S COMMENTS

The candidate received five marks out of five.

In question (a)(i), the candidate recorded correct data in the table, and all data were shown as whole numbers.

The candidate received full marks for accuracy.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

Repeat steps 2 to 10 with different volumes of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ as shown in Table 2.1.

Table 2.1

experiment number	volume of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ / cm^3	volume of $\text{H}_2\text{O}(\text{l})$ / cm^3	volume of $\text{HCl}(\text{aq})$ / cm^3	time / s	rate of reaction / s^{-1}
1	40.0	10.0	5.0	13.35 ✓	74.81 X
2	30.0	20.0	5.0	18.32 ✓	54.59
3	25.0	25.0	5.0	24.53 X	40.77
4	20.0	30.0	5.0	27.84	35.92
5	15.0	35.0	5.0	39.28	25.46
6	10.0	40.0	5.0	55.38	18.06

[5]

EXAMINER'S COMMENTS

The candidate received three marks.

In question (a)(i), the candidate recorded correct data in the table, however all data were shown as incorrect numbers. The candidate received two marks for accuracy.

SAMPLE OF CANDIDATE RESPONSE GRADED “E”

Repeat steps 2 to 10 with different volumes of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ as shown in Table 2.1.

Table 2.1

experiment number	volume of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ / cm^3	volume of $\text{H}_2\text{O}(\text{l})$ / cm^3	volume of $\text{HCl}(\text{aq})$ / cm^3	time / s	rate of reaction / s^{-1}
1	40.0	10.0	5.0	30,60	33,0
2	30.0	20.0	5.0	35,85	28,0
3	25.0	25.0	5.0	47,00	21,0
4	20.0	30.0	5.0	61,30	16,0
5	15.0	35.0	5.0	90,95	11,0
6	10.0	40.0	5.0	151,00	6,0

for Pedagogical Measurements

[5]

EXAMINER'S COMMENTS

The candidate received two marks.

In question (a)(i), the candidate recorded correct data in the table, however all data were shown as incorrect numbers. The candidate received one mark for accuracy.

QUESTION 2 a (ii), b(i)

2 a (ii) The rate of reaction is calculated using this expression.

$$\text{Rate of reaction} = \frac{1000}{\text{time}}$$

Calculate the rate of reaction for experiments 1 to 6.

Record your results in Table 2.1 to **two** significant figures.

[1]

- (b) (i)** Calculate the values of the log(volume of Na₂S₂O₃(aq)) and log(rate of reaction) and write them in Table 2.2.

Table 2.2

experiment number	<u>log</u> (volume of Na ₂ S ₂ O ₃ (aq))	<u>log</u> (rate of reaction)
1		
2		
3		
4		
5		
6		

[2]

MARK SCHEME

2a(ii)	All rates correctly calculated and recorded to 2 significant figures.	[1]	
2b(i)	all log values of Na ₂ S ₂ O ₃ (aq) correctly calculated.	1	
	all log values of rate of reaction correctly calculated	1 [2]	

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

- (ii) The rate of reaction is calculated using this expression.

$$\text{Rate of reaction} = \frac{1000}{\text{time}}$$

Calculate the rate of reaction for experiments 1 to 6.

Record your results in Table 2.1 to **two** significant figures.

[1]

- (b) (i) Calculate the values of the $\log(\text{volume of Na}_2\text{S}_2\text{O}_3(\text{aq}))$ and $\log(\text{rate of reaction})$ and write them in Table 2.2.

Table 2.2

experiment number	$\log(\text{volume of Na}_2\text{S}_2\text{O}_3(\text{aq}))$	$\log(\text{rate of reaction})$
1	1.0 1.6	1.7
2	1.0 1.5	1.6
3	1.0 1.4	1.5
4	1.0 1.3	1.4
5	1.0 1.2	1.2
6	1.0 1.0	1.0

[2]

EXAMINER'S COMMENTS

The candidate received three marks.

In question (a)(ii), the candidate got one mark for correctly calculating all rates and recording the results to 2 significant figures.

In question (b)(i), the candidate received two marks for correctly calculating all log volumes, and also correctly finding out all log values of rate of reaction.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

(ii) The rate of reaction is calculated using this expression.

$$\text{Rate of reaction} = \frac{1000}{\text{time}}$$

Calculate the rate of reaction for experiments 1 to 6.

Record your results in Table 2.1 to **two** significant figures.

[1]

(b) (i) Calculate the values of the log(volume of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$) and log(rate of reaction) and write them in Table 2.2.

Table 2.2

experiment number	log(volume of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$)	log(rate of reaction)
1	1.60	1.87
2	1.48	1.74
3	1.40	1.61
4	1.30	1.56
5	1.18	1.44
6	1.00	1.26

[2]

EXAMINER'S COMMENTS

The candidate received two marks.

In question (a)(ii), the candidate correctly calculated all rates, however they lost one mark for incorrectly writing the significant figures in the answer.

In question (b)(i), the candidate correctly calculated all log volumes, and they also correctly found out all log values of the rate of reaction.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

- (ii) The rate of reaction is calculated using this expression.

$$\text{Rate of reaction} = \frac{1000}{\text{time}}$$

Calculate the rate of reaction for experiments 1 to 6.

Record your results in Table 2.1 to **two** significant figures.

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[1]

- (b) (i) Calculate the values of the $\log(\text{volume of Na}_2\text{S}_2\text{O}_3(\text{aq}))$ and $\log(\text{rate of reaction})$ and write them in Table 2.2.

Table 2.2

experiment number	$\log(\text{volume of Na}_2\text{S}_2\text{O}_3(\text{aq}))$	$\log(\text{rate of reaction})$
1	-1.398	1.877
2	-1.523	1.793
3	-1.602	1.734
4	-1.699	1.691
5	-1.824	1.534
6	-2	1.356

[2]

EXAMINER'S COMMENTS

The candidate received two marks.

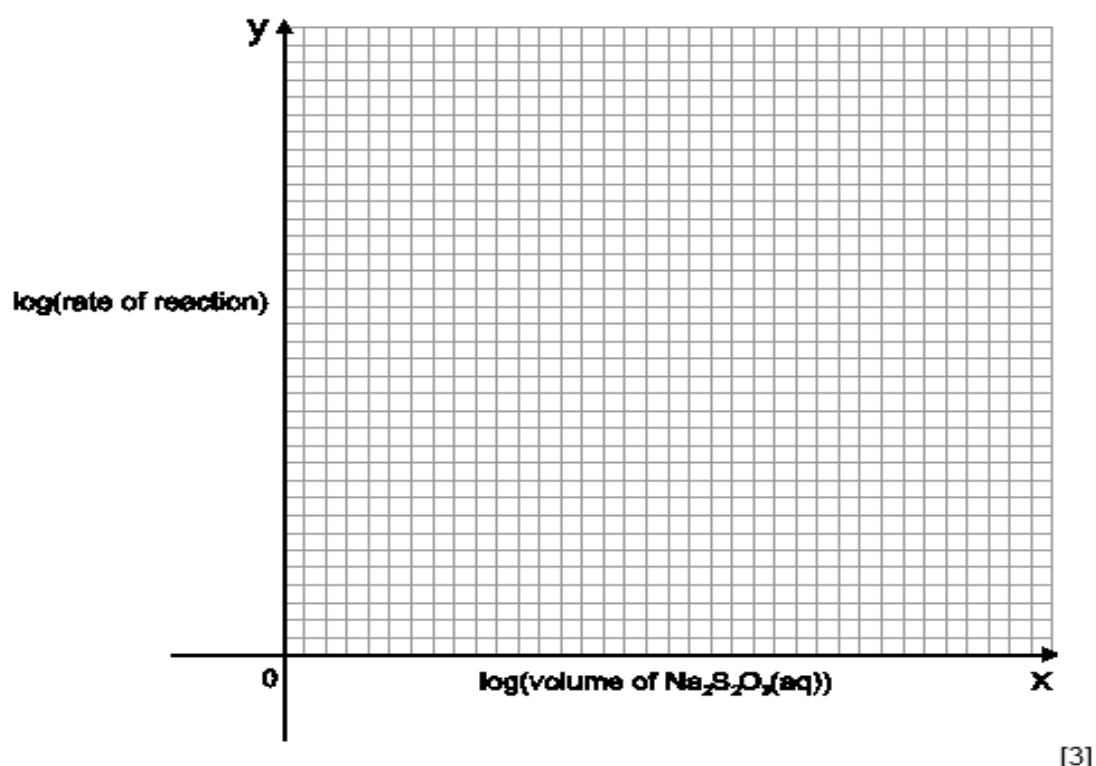
In question (a)(ii), the candidate correctly calculated all rates; however, they lost one mark for incorrectly writing the significant figures in the answer.

In question (b)(i), the candidate lost one mark because all log volumes were incorrectly calculated. However, they received one mark for correctly calculating all log values of the rate of reaction.

QUESTION 2 b (ii),(iii)

2 b (ii) Plot $\log(\text{rate of reaction})$ against the $\log(\text{volume of Na}_2\text{S}_2\text{O}_3(\text{aq}))$.

Draw a line of best fit.



(iii) The order of reaction with respect to $\text{Na}_2\text{S}_2\text{O}_3$ is the gradient of this graph.

Use your graph to suggest the order of reaction with respect to $\text{Na}_2\text{S}_2\text{O}_3$.

[1]

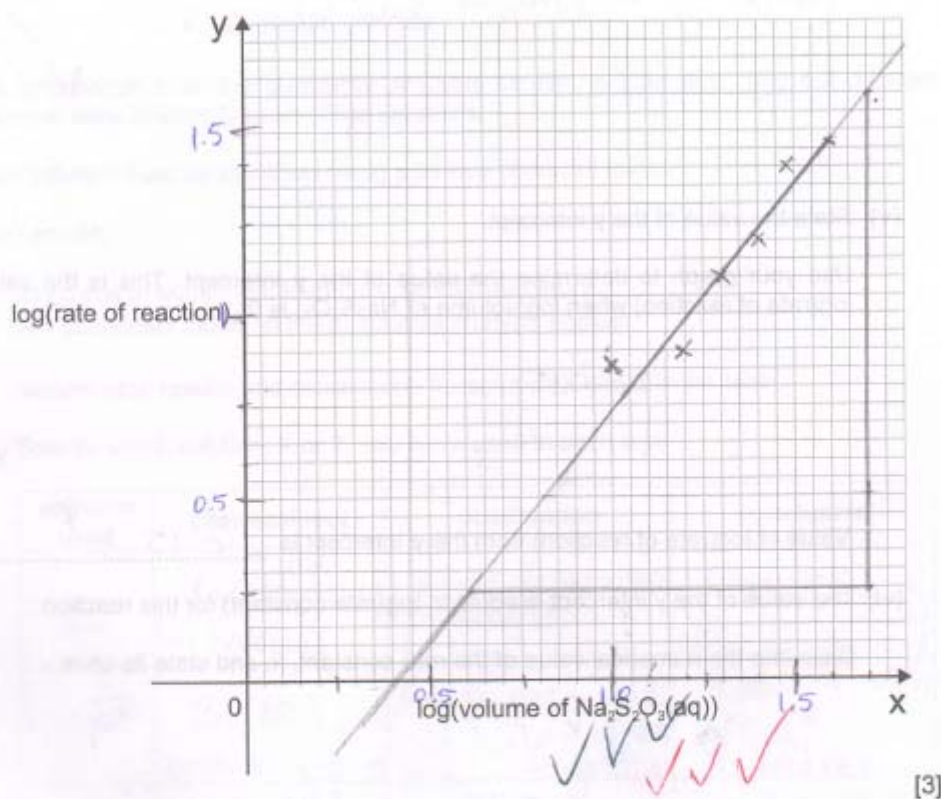
MARK SCHEME

2b(ii)	All points correctly calculated and plotted to within $\frac{1}{2}$ small square	1	
	Use approximately part $\frac{2}{3}$ of area for graph	1	
	Line of best fit	1	
		[3]	
2b(iii)	First order with evidence that the gradient is 1	[1]	

SAMPLE OF CANDIDATE RESPONSE GRADED “A”

(ii) Plot $\log(\text{rate of reaction})$ against the $\log(\text{volume of Na}_2\text{S}_2\text{O}_3(\text{aq}))$.

Draw a line of best fit.



(iii) The order of reaction with respect to $\text{Na}_2\text{S}_2\text{O}_3$ is the gradient of this graph.

Use your graph to suggest the order of reaction with respect to $\text{Na}_2\text{S}_2\text{O}_3$.

$$\frac{\Delta y}{\Delta x} = \frac{1.6 - 0.25}{1.7 - 0.6} \approx 1.22 \text{ which is approximately } 1$$

first order. ✓ ✓

EXAMINER'S COMMENTS

The candidate received four marks.

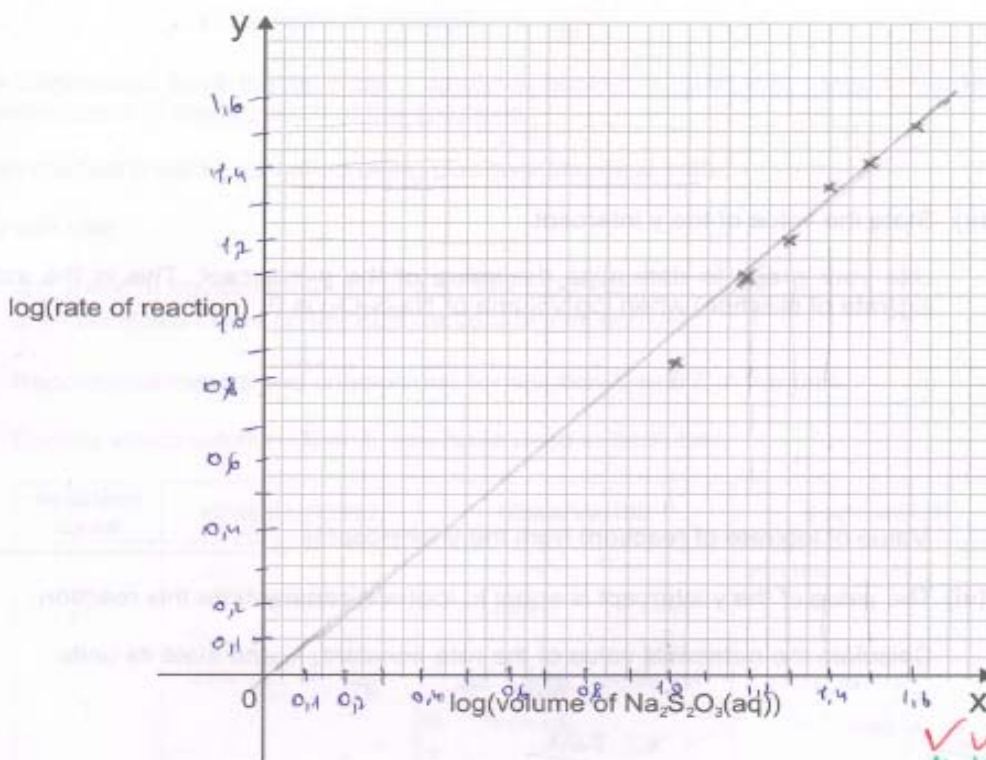
In question (b)(ii), the candidate got all three marks for correctly calculating points, plotting them with the correct size, and drawing the line of best fit accurately.

In question 2b (iii), the candidate correctly suggested the order with evidence.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

(ii) Plot $\log(\text{rate of reaction})$ against the $\log(\text{volume of Na}_2\text{S}_2\text{O}_3(\text{aq}))$.

Draw a line of best fit.



(iii) The order of reaction with respect to $\text{Na}_2\text{S}_2\text{O}_3$ is the gradient of this graph.

Use your graph to suggest the order of reaction with respect to $\text{Na}_2\text{S}_2\text{O}_3$.

First order

✓✓✓✓ 3
[3]

✗ 0
[1]

EXAMINER'S COMMENTS

The candidate received three marks.

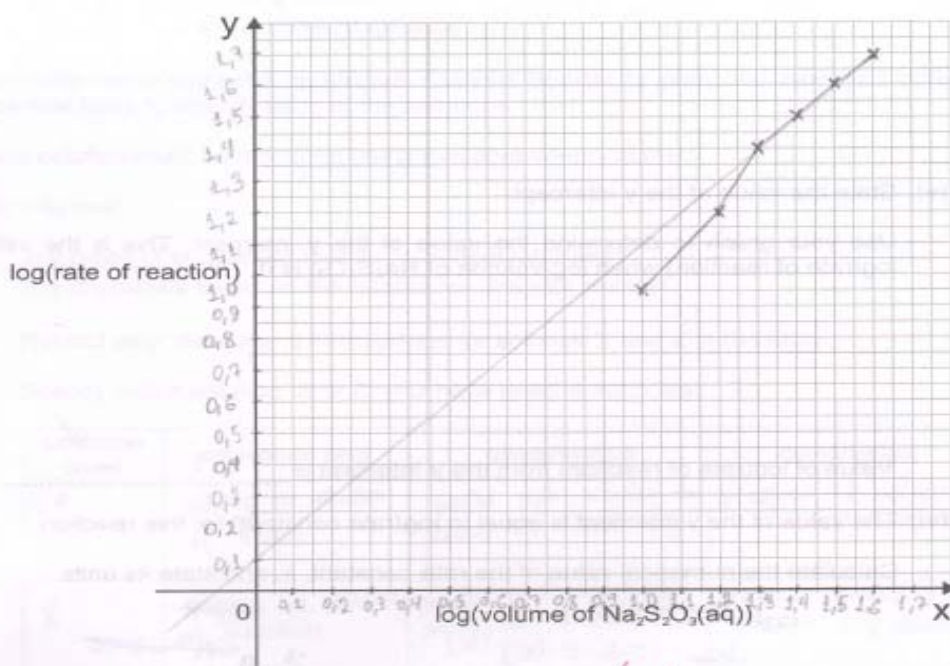
In question (b)(ii), the candidate got all three marks for correctly calculating points and plotting them with the correct size. The line of best fit was also drawn correctly.

In question (b)(iii), the candidate lost one mark for suggesting the order of reaction without providing evidence.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

(ii) Plot $\log(\text{rate of reaction})$ against the $\log(\text{volume of Na}_2\text{S}_2\text{O}_3(\text{aq}))$.

Draw a line of best fit.



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[3]

(iii) The order of reaction with respect to $\text{Na}_2\text{S}_2\text{O}_3$ is the gradient of this graph.

Use your graph to suggest the order of reaction with respect to $\text{Na}_2\text{S}_2\text{O}_3$.

1st order

[1]

EXAMINER'S COMMENTS

The candidate received two marks.

In question (b)(ii), the candidate got two marks for correctly calculating points and plotting them with the correct size. However, the candidate lost one mark for an incorrect drawn line of best fit.

In question (b)(iii), the candidate lost one mark for suggesting the order of reaction without providing evidence.

QUESTION 2 b (iv), (v), (vi)

2 b (iv) Write the rate equation for this reaction, when $[HCl]$ is kept constant.

[1]

(v) State the value of the **y** intercept.

Use your graph to determine the value of the **y** intercept. This is the value of log(rate of reaction) when log(volume of $Na_2S_2O_3$) is 0.

Value of log(rate of reaction) from the **y** intercept is

[1]

(vi) The value of the **y** intercept is equal to log(rate constant) for this reaction.

Calculate the numerical value of the rate constant, k , and state its units.

numerical value

unit

[1]

MARK SCHEME

2b(iv)	Rate = $k [S_2O_3^{2-}]$	[1]	Accept ecf from b(iii)
2(b)(v)	correct answer from graph	[1]	
2(b)(vi)	correct value of k calculated from 2(b)(v) and s^{-1}	[1]	Both needed Accept if student used given rate of reaction unit $dm^3 mol^{-1} s^{-1}$

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

(iv) Write the rate equation for this reaction, when $[HCl]$ is kept constant. $r = k [Na_2S_2O_3]$ [1]	For Examiner Use ✓ 1
(v) State the value of the y intercept. Use your graph to determine the value of the y intercept. This is the value of $\log(\text{rate of reaction})$ when $\log(\text{volume of } Na_2S_2O_3)$ is 0. Value of $\log(\text{rate of reaction})$ from the y intercept is 0.2 [1]	1
(vi) The value of the y intercept is equal to $\log(\text{rate constant})$ for this reaction. Calculate the numerical value of the rate constant, k, and state its units. $k = 10^{0.2} = 1.58 \text{ mol dm}^{-3} \text{ s}^{-1}$ numerical value 1.58 unit s^{-1} [1]	1

EXAMINER'S COMMENTS

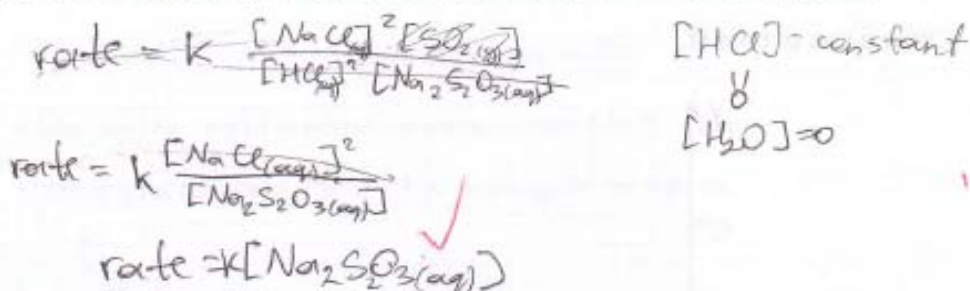
The candidate received three marks out of three.

In question (b)(iv), the candidate got one mark for correctly writing the rate equation.

In (b)(v), the candidate received one mark for correctly identifying the value of the intercept.

In question (b)(vi), the candidate got one mark for correctly calculating the value of the rate constant and for correctly stating the units.

(iv) Write the rate equation for this reaction, when $[HCl]$ is kept constant.



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1

(v) State the value of the y intercept.

Use your graph to determine the value of the y intercept. This is the value of $\log(\text{rate of reaction})$ when $\log(\text{volume of } Na_2S_2O_3)$ is 0.

$$y = -0,35$$

1

Value of $\log(\text{rate of reaction})$ from the y intercept is $-0,35$ [1]

(vi) The value of the y intercept is equal to $\log(\text{rate constant})$ for this reaction.

Calculate the numerical value of the rate constant, k , and state its units.

$$k = \frac{\text{rate}}{[Na_2S_2O_3(aq)]}$$

$$k = 10^{+0,35} = 2,24 \frac{\text{dm}^3}{\text{mol} \cdot \text{s}}$$

numerical value $2,24$ unit $\frac{\text{dm}^3}{\text{mol} \cdot \text{s}}$ [1]

0

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

EXAMINER'S COMMENTS

The candidate received three marks out of three.

In question (b)(iv), the candidate got one mark for correctly writing the rate equation.

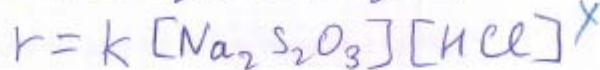
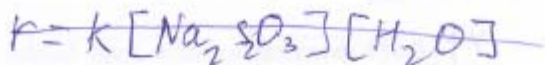
In question (b)(v), the candidate received one mark for correctly identifying the value of the intercept.

In question (b)(vi), the candidate lost one mark for incorrectly calculating the value of the rate constant.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

(iv) Write the rate equation for this reaction, when [HCl] is kept constant.

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Use



[1]

(v) State the value of the y intercept.

Use your graph to determine the value of the y intercept. This is the value of log(rate of reaction) when log(volume of $\text{Na}_2\text{S}_2\text{O}_3$) is 0.

0 ✓ ✓

Value of log(rate of reaction) from the y intercept is [1]

(vi) The value of the y intercept is equal to log(rate constant) for this reaction.

Calculate the numerical value of the rate constant, k , and state its units.

$$k = \frac{r}{[\text{Na}_2\text{S}_2\text{O}_3][\text{HCl}]} = 0,2 = \frac{\text{s}^{-1}}{\text{dm}^3 \cdot \text{dm}^3} = \frac{\text{s}^{-1} \cdot \text{cm}^3 \cdot \text{cm}^3}{\text{dm}^3 \cdot \text{dm}^3} = 20000$$

numerical value 0,2 unit $\text{cm}^6 \text{s}^{-1}$ ✓ X [1]

EXAMINER'S COMMENTS

The candidate received one mark.

In question (b)(iv), the candidate lost one mark for an incorrectly written rate equation.

In question (b)(v), the candidate received one mark for correctly identifying the value of the intercept.

In question (b)(vi), the candidate lost one mark for an incorrectly calculated value of the rate constant and for incorrectly stating its units.

QUESTION 2 c (i)

2 (c) You are provided with samples of two colourless solutions, **X** and **Z**.

These solutions are:

- zinc sulfate
- aluminium chloride.

Use information from the qualitative analysis booklet to plan and carry out different chemical tests to identify each of the solutions.

Each solution must be identified using **positive** chemical tests.

You can use:

- any piece of equipment found in a school laboratory;
- any chemicals found on the qualitative analysis booklet.

(i) Record your results and conclusions for solution **X** and **Z** in the table.

Specify which solution, **X** or **Z**, you have used in each test.

solution used	chemical test	observation	conclusion

[6]

MARK SCHEME

2(c)(i)	chemical test	observation with X	conclusion	[3]	Ignore Other test and other results
	use NaOH(aq) test and uses Ba ²⁺ (aq) test (1)	white ppt that soluble in excess and white ppt insoluble (1)	zinc and sulfate present (1)		
2(c)(ii)	chemical test	observation with Z	conclusion	[3]	Ignore Other test and other results
	Use NH ₃ (aq) and Use Ag ⁺ (aq) test (1)	white ppt insoluble in excess (1) and white ppt (1)	aluminium and chloride present (1)		

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

(c) You are provided with samples of two colourless solutions, X and Z.

These solutions are:

- zinc sulfate
- aluminium chloride

Use information from the qualitative analysis booklet to plan and carry out different chemical tests to identify each of the solutions.

Each solution must be identified using positive chemical tests.

You can use:

- any piece of equipment found in a school laboratory;
- any chemicals found on the qualitative analysis booklet.

(i) Record your results and conclusions for solution X and Z in the table.

Specify which solution, X or Z, you have used in each test.

solution used	chemical test	observation	conclusion
X	addition of $\text{Ba}(\text{NO}_3)_2$ and excess HNO_3	white ppt insoluble in excess dilute strong acid	SO_4^{2-} anions are in solution X - zinc sulphate
Z	addition of AgNO_3 (Ag^+) and NH_3	white ppt that dissolved in excess ammonia	Cl^- anions are in solution Z - aluminium chloride
X	addition of NH_3	white ppt dissolved in excess NH_3	Zn^{2+} cations are in solution Zinc chloride
Z	addition of NH_3	white ppt insoluble in excess NH_3	X - zinc chloride

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EXAMINER'S COMMENTS

The candidate received six marks out of six.

In question (c)(i), the candidate got all six marks for providing a complete qualitative analysis plan, including positive tests for each of the cations and anions with appropriate observations. Consequently, they drew correct conclusions.

SAMPLE OF CANDIDATE RESPONSE GRADED “C”

(c) You are provided with samples of two colourless solutions, X and Z.

These solutions are:

- zinc sulfate
- aluminium chloride.

Use information from the qualitative analysis booklet to plan and carry out different chemical tests to identify each of the solutions.

Each solution must be identified using **positive** chemical tests.

You can use:

- any piece of equipment found in a school laboratory;
- any chemicals found on the qualitative analysis booklet.

(i) Record your results and conclusions for solution X and Z in the table.

Specify which solution, X or Z, you have used in each test.

solution used	chemical test	observation	conclusion
X	Addition of Ba^{2+} ($\text{Ba}(\text{NO}_3)_2$) ✓	white ppt	has sulfate ions
X Z	Addition of excess NH_3 ✓	insoluble white ppt	has aluminium ions
Z	Addition of AgNO_3 ✓	white ppt	has chloride ions ✓

[6]

4

X X

EXAMINER'S COMMENTS

The candidate received four marks out of six.

In question (c)(i), the candidate got four marks for providing an incomplete qualitative analysis plan for positive tests for cations and anions with appropriate observations. Consequently, their conclusions were also incomplete and partially correct.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

(c) You are provided with samples of two colourless solutions, X and Z.

These solutions are:

- zinc sulfate
- aluminium chloride.

Use information from the qualitative analysis booklet to plan and carry out different chemical tests to identify each of the solutions.

Each solution must be identified using **positive** chemical tests.

You can use:

- any piece of equipment found in a school laboratory;
- any chemicals found on the qualitative analysis booklet.

(i) Record your results and conclusions for solution X and Z in the table.

Specify which solution, X or Z, you have used in each test.

solution used	chemical test	observation	conclusion
X	NaOH NaOH	no change white ppt insoluble in excess.	cannot determine.
Z	NH_3	✓ white ppt insoluble in excess.	have cation Al^{3+} , Cl^- ✓
X	AgNO_3	✓ white ppt, soluble in excess.	have cations of Zn^{2+}

[6]

X X X

3

EXAMINER'S COMMENTS

The candidate received three marks.

In question (c)(i), the candidate lost three marks for providing an incomplete qualitative analysis plan for positive tests for cations and anions with appropriate observations. Consequently, their conclusions were incomplete and partially correct.